LaTeX Math Style Guide

Ivan Jacob Agaloos Pesigan

One equation should render the same everywhere it is pasted: LaTeX (pdfLaTeX), the Rd PDF manual, pkgdown (KaTeX), Quarto and R Markdown (MathJax), and Beamer. Agents follow `.setup/ai/style/latex-math.md`, which follows this guide and is changed together with it.

1 Form

What a symbol means can change from paper to paper; each project lists its symbols in `.setup/ai/project.md`. Its form does not change, and follows the conventions of linear algebra, statistics, and calculus.

- Scalars and elements: plain italic ($a_{12,i}$, α).
- Vectors: bold lowercase ($\boldsymbol{y}$, $\boldsymbol{\eta}$).
- Matrices: bold uppercase ($\boldsymbol{B}$, $\boldsymbol{\Phi}$).
- Named exception: the Wiener process $\boldsymbol{W}$ is a vector in bold uppercase, the usual form for a stochastic process.

2 Bold

- `\boldsymbol{}` for every vector and matrix, Latin and Greek.
- Do not write `\mathbf` or `\bm` in sources. MathJax has no `\bm`, and `\mathbf` prints upright on the web but italic in the PDF. The preamble keeps its remap to `\bm` for better PDF glyphs, so this rule moves no glyph in a PDF.

- Scalars and elements stay plain italic ($a_{12,i}$).

```
\boldsymbol{x}_{i, t}  \boldsymbol{A}  \boldsymbol{\Sigma}  a_{12, i}
```

3 Symbols

- Transpose: `^{\prime}` (not `^T`, `^{\top}`, or `'`).
- Operators: `\operatorname{}` (`vec`, `vech`, `tr`, `diag`, `Cov`), never `\text` or `\mathrm`.
- Subscripts in braces, with a space after commas: `\boldsymbol{x}_{i, t}`.
- `\left( \right)` around groups of more than one symbol, and `\!` before the `\left(` of a function.
- `\mathbb{E}` and `\mathbb{V}` for expectation and variance.
- The differential is upright, as in ISO 80000-2, with a thin space before it: `\, \mathrm{d} t`, and `\frac{\mathrm{d} y}{\mathrm{d} t}` for a derivative. This is the exception to the operator rule. The partial derivative stays `\partial`.

```
\mathcal{N} \! \left( \boldsymbol{0}, \boldsymbol{\Xi} \right)
\operatorname{vec} \! \left( \boldsymbol{A} \right)
\boldsymbol{A}^{\prime}
```

4 Display Equations

- `equation` when the equation is referred to, `equation*` otherwise.
- A short equation stays on one line.
- A long one breaks before `=`, `+`, and `\sim`, and at commas between parts: one piece per line, indented with a tab.

```
\begin{equation}
        \boldsymbol{\eta}_{i, t}
        =
        \boldsymbol{\alpha}
        + \boldsymbol{B} \boldsymbol{\eta}_{i, t - 1}
        + \boldsymbol{\zeta}_{i, t},
        \quad
        \boldsymbol{\zeta}_{i, t}
        \sim \mathcal{N} \! \left( \boldsymbol{0}, \boldsymbol{\Psi} \right)
        \label{eq:template-var}
\end{equation}
```

5 Macros

None. Plain, portable commands only. Do not use `\newcommand`, `\renewcommand`, or `\DeclareMathOperator` outside `.setup/latex/preamble`.

6 Portable Math

Use only commands that work in all of the contexts above: the Greek and Latin letters that LaTeX defines, `\boldsymbol`, `\mathcal`, `\mathbb`, `\mathrm{d}`, `\operatorname`, `\frac`, `\sqrt`, `\exp`, `\log`, `\sum`, `\prod`, `\int`, `\sim`, `\mid`, `\ldots`, `\left`, `\right`, `\quad`, `\,`, `\hat`, `\tilde`, `\bar`, and the environments `pmatrix` and `aligned`.

Not this	This
`\Beta`, `\Alpha` (not LaTeX)	`\mathrm{B}`, `\mathrm{A}`
`\argmax`, `\argmin`	`\operatorname*{arg\,max}`
`\bm`, `\mathbf`	`\boldsymbol`

rProject's test page (`.setup/quarto/math/math.qmd`) renders these commands with MathJax, and this guide renders them in the PDF, so a change in a renderer shows up.

7 Shared Notation

This table holds the default symbols across projects. A paper can give a symbol another meaning: its `.setup/ai/project.md` ("Models and notation") lists its own symbols and any departures, in the same form. Existing packages and papers keep their symbols until they are next reused. Elements of a vector or matrix use the plain letter. The subscripts i and t are the person and the time.

Symbol	Source	Meaning
Measurement		
`\boldsymbol{y}_{i, t}`	$\boldsymbol{y}_{i,t}$	observed variables
`\boldsymbol{\eta}_{i, t}`	$\boldsymbol{\eta}_{i,t}$	latent variables
`\boldsymbol{\nu}`	$\boldsymbol{\nu}$	measurement intercepts
`\boldsymbol{\Lambda}`	$\boldsymbol{\Lambda}$	factor loadings
`\boldsymbol{\varepsilon}_{i, t}`	$\boldsymbol{\varepsilon}_{i,t}$	measurement errors
`\boldsymbol{\Theta}`	$\boldsymbol{\Theta}$	covariance of the measurement errors
Discrete time (VAR)		
`\boldsymbol{\alpha}`	$\boldsymbol{\alpha}$	intercepts
`\boldsymbol{B}`	$\boldsymbol{B}$	transition matrix
`\boldsymbol{\zeta}_{i, t}`	$\boldsymbol{\zeta}_{i,t}$	process noise
`\boldsymbol{\Psi}`	$\boldsymbol{\Psi}$	covariance of the process noise
Continuous time		
`\boldsymbol{\iota}`	$\boldsymbol{\iota}$	intercepts (linear SDE)
`\boldsymbol{\mu}`	$\boldsymbol{\mu}$	set point (Ornstein-Uhlenbeck)
`\boldsymbol{\Phi}`	$\boldsymbol{\Phi}$	drift matrix
`\boldsymbol{\Sigma}`	$\boldsymbol{\Sigma}$	diffusion covariance
`\boldsymbol{W}_{i, t}`	$\boldsymbol{W}_{i,t}$	Wiener process
`\Delta t`	Δt	time interval
Continuous-time mediation		
`\exp \!  \left( \Delta t \boldsymbol{\Phi} \right)`	$\exp(\Delta t \boldsymbol{\Phi})$	total effects
`\boldsymbol{D}_{\boldsymbol{m}}`	$\boldsymbol{D}_{\boldsymbol{m}}$	removes the mediators, for the direct effects
Cross-sectional mediation		
`X`, `M`, `Y`	X, M, Y	predictor, mediator, outcome
`\alpha`, `\beta`	α, β	paths $X \rightarrow M$, $M \rightarrow Y$
`\tau^{\prime}`	τ'	direct effect

The linear stochastic differential equation and the Ornstein-Uhlenbeck model are

```
\mathrm{d} \boldsymbol{\eta}_{i, t}
=
\left( \boldsymbol{\iota} + \boldsymbol{\Phi} \boldsymbol{\eta}_{i, t} \right) \, \mathrm{d} t
+ \boldsymbol{\Sigma}^{\frac{1}{2}} \, \mathrm{d} \boldsymbol{W}_{i, t}


\mathrm{d} \boldsymbol{\eta}_{i, t}
=
\boldsymbol{\Phi} \left( \boldsymbol{\eta}_{i, t} - \boldsymbol{\mu} \right) \, \mathrm{d} t
+ \boldsymbol{\Sigma}^{\frac{1}{2}} \, \mathrm{d} \boldsymbol{W}_{i, t}
```

8 Tools

`make lint` reports (never fails on) bold, portable math, macros, and symbols, in LaTeX sources, roxygen `\eqn{}` and `\deqn{}`, and Markdown math.

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/